



Impacts of regime shift on the fishery ecosystem in the coastal area of Kyoto prefecture, Sea of Japan, assessed using the Ecopath model

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Abstract

The present study assessed the impact of regime shifts on the ecosystems that support fisheries in the coastal area of Kyoto Prefecture, the Sea of Japan. Ecopath models were constructed during cold (1985) and warm (2013) regime periods, when the fishery structures were different. Both models evaluated by pre-balance diagnostics were reliably constructed. Multiple ecosystem indicator values were compared between the two periods to assess changes in the ecosystem structure and the impact of fisheries on the ecosystem. The major change in catches in the coastal area of Kyoto Prefecture between 1985 and 2013 was that the Japanese sardine drastically declined and Spanish mackerel increased. The primary production requirements of fisheries (%PPR) were higher in 1985, reflecting the greater impact of sardine. The mean trophic level of catch and system omnivory index suggested that stability of trophic network increased after the sharp decline in sardine biomass. The %PPR, the loss of secondary production due to fisheries exploitation and probability analysis of fishery sustainability suggested that the fishery structure in 2013, which was dominated by set nets, contributed to a more sustainable fishery than the structure in 1985, when fishery was dominated by purse seines and set nets.

Keywords Ecopath · Regime shift · Sustainable fishery

Introduction

Fluctuations in fishery resources are strongly influenced by both fishery catch and the marine environment. Decadal-scale variability in the abundance of prominent species of small pelagic fishes in the North Pacific Ocean has been reported to be generally associated with climatic/oceanographic variability, which is referred to as a regime shift (Chavez et al. 2003; Yatsu 2019). Ecological regime shifts can be defined as abrupt changes at several trophic levels, leading to rapid ecosystem reconfiguration between alternative states (Andersen et al. 2009). Regime shifts can be caused by various stressors, among which temperature appears to be an important factor related to changes in marine ecosystems (Takasuka et al. 2008). Understanding ecosystem structures and functions to evaluate and manage both ecosystems and fisheries comprehensively and concomitantly is essential to develop an effective approach for sustainable fisheries (Costello 2012; Piet et al. 2017).

Ecosystem-based management (EBM) has attracted considerable attention as an effective approach for multi-species fishery management (Hilborn and Ovando 2014; Dolan et al.

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2015). To date, the management of fishery resources in Japan has mainly adopted a single-species (or single-population) management policy based on total allowable catch (TAC) in conjunction with traditional effort management (Makino 2011; Ichinokawa et al. 2017). Currently, more than 80% of the total Japanese catch is from multi-species fisheries, such as set nets (set net refers to stationary uncovered pound net in Japan), bottom trawl, and purse seine fisheries (Ministry of Agriculture, Forestry and Fisheries 2020); therefore, the application of EBM should become an essential and powerful tool for the management of these multi-species fisheries.

Ecosystem models that can handle multiple stocks in consideration of energy and/or organic matter flow through predator–prey interactions have been widely used in recent decades. Among them, the Ecopath model was developed in 1984 and is currently one of the most used ecosystem models in the world (over 900 published models as of August 2020), with many plugin functions developed and added (Polovina 1984; Colléter et al. 2013). Ecopath is a multi-species ecosystem model that can be used to analyze the structure and function of marine ecosystems and evaluate the effects of fisheries (Christensen and Walters 2004). The European Commission recently adopted Ecopath in its management strategy to assess the health of ecosystems and fisheries in the North Sea (Mackinson et al. 2018). It is expected that the Ecopath model can be used to examine changes in ecosystem structure and the impact of fishing on the ecosystem caused by regime shifts. Analysis of the effects of regime shift can provide novel insights into the climate–ecosystem relationship and thus impact fisheries in the future.

This study aimed to assess the impact of regime shifts on fishery ecosystems, the ecosystems that support fisheries, in the coastal area of Kyoto Prefecture (CAK), the Sea of

Japan (Fig. 1). The CAK is one of ideal areas in Japan for ecosystem modeling because of the presence of substantial detailed sampling and fishery data (Online Resource 1, Fig. S2). The annual total catch of Kyoto Prefecture was approximately 40,000 tons in the 1970s, but it increased sharply in the 1980s and reached a peak of 106,000 tons in 1988. However, catches have followed a long-term declining trend since then, reaching 11,000 tons in 2018. Fluctuations in total catch was mainly caused by a sharp increase in catches of the sardine *Sardinops melanostictus* until 1988 (87,000 tons) and a sharp decrease after that (11 tons in 2002) (Fig. 2c). In the Sea of Japan, the shift from a cold regime to a warm regime that occurred in the late 1980s had a substantial influence on the fluctuation of fish communities (Tian 2014; Tian et al. 2014). There is a positive correlation between sardine stock size and sea surface temperature (SST) in the Kuroshio Extension region (Noto and Yasuda 1999). Yatsu (2019) reported that the biomass of the Tsushima Current stock of the sardine has continuously increased since the early 1970s owing to a high recruitment per spawning stock and low exploitation rate until the 1988/89 regime shift. Since the 1988/1989 regime shift, poor ocean productivity has been observed, and dominant small pelagic fish have shifted from sardine to the Japanese anchovy *Engraulis japonicus* (Yatsu 2019). Similar biomass fluctuations for sardine and anchovy in the Sea of Japan has been confirmed (Ohshimo et al. 2009). This structural ecosystem change is likely to have affected sardine catch in CAK. In Kyoto Prefecture, the main fishing methods were purse seines and set nets before the regime shift, but set nets have become the primary fishing method after it (Online Resource 2, Fig. S2) (Kyoto Prefectural Fishery Cooperative Association 2015). The set net has been widely used in Japan, accounting for

Fig. 1 Study area, the coastal area of Kyoto Prefecture (CAK), Japan. The model area was from the coastal line to a bathymetry depth of 240 m, covering 2230 km². The dashed line shows the border of the model area. White circles show the locations of beam trawl sampling at depths of 5, 10, 30, 60, 100, 150, and 200 m

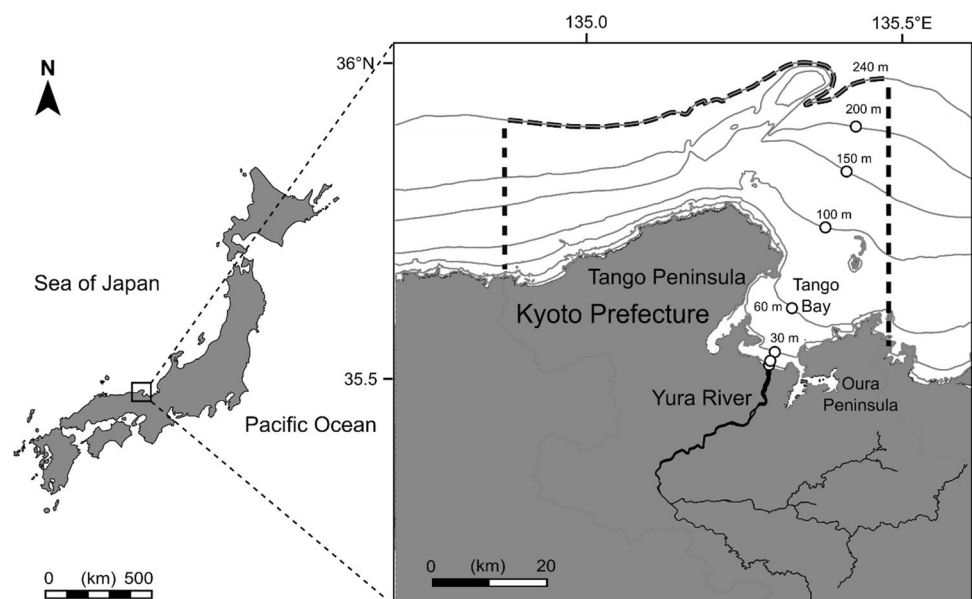
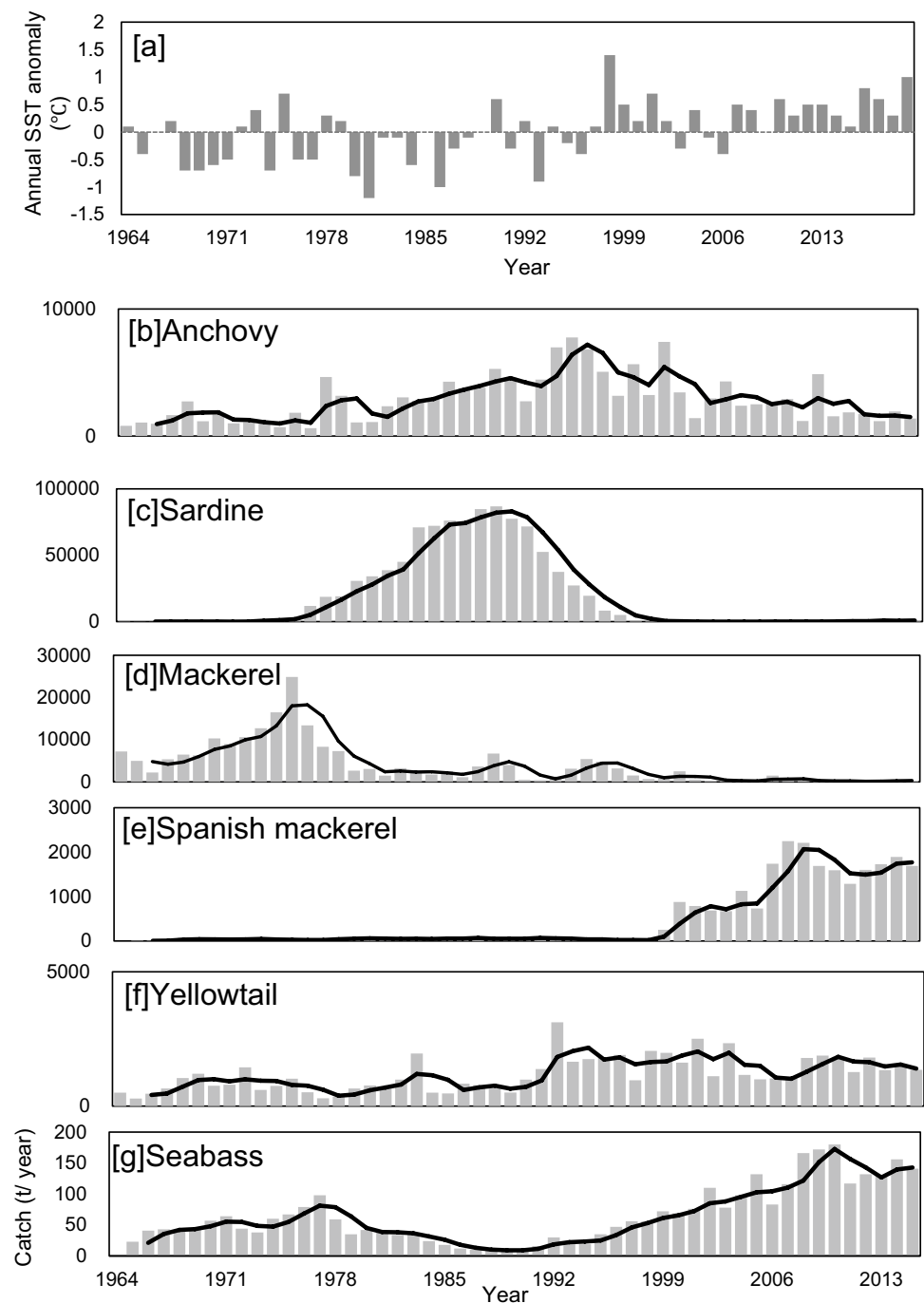


Fig. 2 **a** Annual sea surface temperature (SST) anomaly in the southwestern Japan Sea from 1964 to 2015. Total commercial catches (bottom 6 panels) in the CAK from 1964 to 2015. The lines indicate moving average (period of 3). The three species (**b** anchovy, **c** sardine, and **d** mackerel) whose catches decreased the most and the three species (**e** Spanish mackerel, **f** yellowtail, and **g** seabass) whose catches increased the most from 1985 to 2013 in the catch data of Kyoto Prefecture



approximately 30% of the nation's total coastal fishery catch in 2016 (Tamaki 2018).

In the present study, we constructed Ecopath models of the CAK for before (model year: 1985) and after (model year: 2013) regime shift periods to understand the change in trophic structure and impacts of fishery catches on marine resources between the two regimes. These two models also indicate

differences between ecosystems under the different major fishing methods, purse seine and set net fisheries in 1985 and mainly set net fisheries in 2013. The model indicators of the effects of fishing on ecosystem functioning were compared between these two regimes, and the sustainability of coastal fisheries in Kyoto Prefecture was evaluated.

Materials and methods

Model area

The study area consisted of coastal waters off the Kyoto Prefecture, from the coastline to the 240 m bathymetry line (approximately 35°9 N latitude) and from 134°8 to 135°5 E longitude. The study area, referred to as CAK, covered 2230 km² (Fig. 1). The coastal area is affected by the Tsushima warm current, and the environment and biological production of Tango Bay are affected by the water from the Yura River (Itoh et al. 2016; Watanabe et al. 2017). The environment, ecosystems, and biological production mechanisms of Tango Bay have been well studied (e.g., Akiyama et al. 2015; Watanabe et al. 2017; Omweri et al. 2018). Many small pelagic fishes, such as the jack mackerel *Trachurus japonicus* and anchovy, and coastal fishes, such as the red sea bream *Pagrus major* and rockfish *Sebastes* spp., seasonally visit or dwell in this area (Masuda et al. 2008; Terashima et al. 2020). Large offshore species, such as the yellowtail *Seriola quinqueradiata* and the Spanish mackerel *Scomberomorus niphonius* also enter Tango Bay for foraging (Terashima et al. 2020).

Ecopath

The Ecopath model was used to construct the 1985 and 2013 model ecosystems and evaluate fishery effects in the CAK. The differences in biomass, production, and throughput between the 1985 and 2013 models were pronounced. In particular, the biomass of sardine in 1985 was 135 times higher than that in 2013. The Ecopath model (Christensen and Pauly 1992; Pauly et al. 2000) creates a static mass-balanced snapshot of the resources in an ecosystem and their interactions and is represented by trophically linked biomass (Christensen et al. 2005). Ecopath estimates trophic mass balance linkages using data on biomass, production ratio, consumption ratio, diet composition, and landings. This relationship can be expressed as:

$$B_i(P/B)_i EE_i - Y_i - E_i - \sum_{j=1}^n B_j(Q/B)_j DC_{ji} = 0 \quad (1)$$

where B_i and B_j are the biomass of the prey functional group i and predator functional group j , respectively; $(P/B)_i$ is the production per biomass ratio of group i , which is equivalent to the total mortality coefficient Z of Allen (1971); EE_i is the ecotrophic efficiency of group i , which is described as the proportion of the production utilized within the system; Y_i is the total catch rate of the functional group i ; E_i is the import/export of group i ; $(Q/B)_j$ is the consumption per biomass of

predator j ; and DC_{ji} is the diet composition for predator j feeding on prey i .

Ecopath requires DC_{ji} and Y_i for all functional groups. Three of the four parameters B_i , $(P/B)_i$, $(Q/B)_i$, and EE_i must be known for each functional group i ; the fourth parameter can be estimated using a system of equations. In this study, Ecopath with Ecosim software (version 6.6) was used to construct the models for CAK.

Selection of functional groups

The fishery resources caught by several types of fishing gear and benthic communities which were untargeted by commercial fisheries were sorted into 40 functional groups (Table 1) representing biological components pooled together according to similarities in species trophic properties and distributions (Coll  ter et al. 2012). The scientific names of each functional group are provided in Online Resource 3 (Table S3). Out of the 40 functional groups used in this study (Table 1), 13 were benthic invertebrates (#22, #23, #24, #25, #26, #27, #28, #32, #33, #34, #35, #36, and #37), 8 were demersal fishes (#11, #12, #13, #15, #18, #29, #30, and #31), 16 were pelagic fishes or cephalopods (#1, #2, #3, #4, #5, #6, #7, #8, #9, #10, #14, #16, #17, #19, #20, and #21), 2 were planktons (#38 and #39), and 1 was a detritus (#40).

In the CAK, functional groups were separated into two coexisting categories: settled functional groups that complete their life cycle in this area and functional groups that migrate outside the model area and temporarily use the CAK. Some large predators only temporarily use the model area as migrants. However, when considering coastal fisheries in Kyoto, these functional groups contribute substantially to the caught biomass and its economic value, making them indispensable.

Input parameters

The basic input parameters, that is, biomass (B), production per biomass (P/B), consumption per biomass (Q/B), diet composition (DC), and landings (Y), for each functional group were obtained from the literature (Table 1, Table 2, and Online Resource 4, Table S4). Benthic organisms, including untargeted species by commercial fisheries, are important prey items for carnivorous animals (Virnstein 1977). Therefore, the benthic community in the CAK was sampled to collect data on species composition and biomass. Detailed methods for functional group selection and biomass estimation for benthos caught by a beam trawl net are presented in Online Resource 1. The annual catch statistics of Kyoto Prefecture in 1985 and 2013 (Online Resource 4, Table S4) were used to calculate the yearly catch of each fishery functional group in the CAK (Kyoto Prefectural

Table 1 Basic output estimates of 40 functional groups in the CAK in 1985 and 2013.

	Functional group	<i>TL</i>	<i>B</i>	<i>P/B</i>	<i>Q/B</i>	<i>P/Q</i>	<i>EE</i>
1	Sardine	3.00	114.92	0.69	5.8	0.12	0.51
2	Round herring	3.00	0.24	3.6	12.75	0.28	0.77
3	Anchovy	3.00	4.36	2.96	11.39	0.26	0.46
4	Mackerel	3.50	4.27	0.66	6.14	0.11	0.78
5	Jack mackerel	3.09	1.45	1.52	5.97	0.25	0.56
6	Yellowtail	3.91	0.68	0.93	4.87	0.19	0.60
7	Frigate tuna	4.01	0.34	1.08	5.61	0.19	0.50
8	Tuna	4.07	0.13	1.08	8.79	0.12	0.46
9	Sailfish	4.08	0.03	1.08	8.44	0.13	0.46
10	Spanish mackerel	3.97	0.09	0.93	6.22	0.15	0.45
11	Stingray	2.89	0.02	0.39	5.01	0.08	0.63
12	Flounder	3.49	0.07	0.56	2.26	0.25	0.70
13	Black porgy	3.29	0.01	0.7	3.55	0.20	0.50
14	Red seabream	3.52	0.05	0.41	2.38	0.17	0.86
15	Tilefish	3.37	0.003	0.7	3.88	0.18	0.50
16	Seabass	3.37	0.02	0.7	4.43	0.16	0.64
17	Flying fish	3.00	1.63	0.69	4.49	0.15	0.45
18	Rockfish	3.51	0.06	0.7	3.81	0.18	0.53
19	Barracuda	3.58	1.18	0.69	4.49	0.15	0.80
20	Flying squid	3.59	0.04	4	25	0.16	0.30
21	Other squids	3.23	0.67	4	25	0.16	0.41
22	Octopus	3.47	0.11	4	25	0.16	0.46
23	Ivory shell	2.16	0.02	1.5	5	0.30	0.00
24	Sea cucumber	2.00	0.35	0.64	2.13	0.30	0.38
25	Other conch	2.00	0.06	1.5	5	0.30	0.68
26	Crab	2.30	0.20	3	10	0.30	0.12
27	Prawn	2.20	0.07	3	10	0.30	0.28
28	Bivalve	2.00	0.05	5	16.67	0.30	0.70
29	Dragonet	2.72	0.0004	2.96	14.73	0.20	0.16
30	Tongue sole	2.71	0.001	2.96	14.08	0.21	0.01
31	Goby	2.55	0.001	3.5	24.46	0.14	0.01
32	Shrimp	2.20	0.08	3	10	0.30	0.51
33	Starfish	2.17	0.14	0.17	0.57	0.30	0.02
34	Brittle star	2.00	29.99	0.2	0.67	0.30	0.00
35	Polychaeta	2.00	9.53	1	3.33	0.30	0.12
36	Hermit crab	2.60	0.04	3	10	0.30	0.02
37	Mysid	2.10	0.01	45	150	0.30	0.10
38	Zooplankton	2.00	15.00	46.21	154.03	0.30	0.86
39	Phytoplankton	1.00	64.83	64.21			0.50
40	Detritus	1.00	43.00				0.10
	Functional group	<i>TL</i>	<i>B</i>	<i>P/B</i>	<i>Q/B</i>	<i>P/Q</i>	<i>EE</i>
1	Sardine	3.00	0.85	0.69	5.8	0.12	0.86
2	Round herring	3.00	0.004	3.6	12.75	0.28	0.85
3	Anchovy	3.00	1.95	2.96	11.39	0.26	0.83
4	Mackerel	3.20	0.13	1.08	6.14	0.18	0.82
5	Jack mackerel	3.17	0.82	2	5.97	0.34	0.87
6	Yellowtail	3.76	1.41	0.93	4.87	0.19	0.42
7	Frigate tuna	3.75	0.15	1.08	5.61	0.19	0.33
8	Tuna	4.08	0.08	1.08	8.79	0.12	0.27

Table 1 (continued)

	Functional group	<i>TL</i>	<i>B</i>	<i>P/B</i>	<i>Q/B</i>	<i>P/Q</i>	<i>EE</i>
9	Sailfish	4.07	0.03	1.08	8.44	0.13	0.27
10	Spanish mackerel	3.49	1.99	0.93	6.22	0.15	0.84
11	Stingray	3.01	0.00002	0.39	5.01	0.08	0.63
12	Flounder	3.34	0.04	0.56	2.26	0.25	0.76
13	Black porgy	3.20	0.03	0.7	3.55	0.20	0.47
14	Red seabream	3.27	0.10	0.41	2.38	0.17	0.81
15	Tilefish	3.03	0.03	0.7	3.88	0.18	0.47
16	Seabass	3.27	0.14	0.7	4.43	0.16	0.68
17	Flying fish	3.00	0.25	0.69	4.49	0.15	0.57
18	Rockfish	3.09	0.05	0.7	3.81	0.18	0.69
19	Barracuda	3.11	0.49	0.69	4.49	0.15	0.69
20	Flying squid	3.10	0.51	4	25	0.16	0.57
21	Other squids	3.04	1.73	4	25	0.16	0.27
22	Octopus	2.43	0.26	4	25	0.16	0.04
23	Ivory shell	2.22	0.00002	1.5	5	0.3	0.16
24	Sea cucumber	2.00	0.20	0.64	2.13	0.3	0.38
25	Other conch	2.00	0.10	1.5	5	0.3	0.68
26	Crab	2.30	0.01	3	10	0.3	0.68
27	Prawn	2.40	0.02	3	10	0.3	0.63
28	Bivalve	2.20	0.42	3	10	0.3	0.64
29	Dragonet	2.81	0.01	2.96	14.73	0.20	0.14
30	Tongue sole	2.71	0.02	2.96	14.08	0.21	0.00
31	Goby	2.92	0.01	2.96	24.46	0.12	0.18
32	Shrimp	2.20	0.14	3	10	0.3	0.71
33	Starfish	2.24	0.07	0.17	0.57	0.3	0.10
34	Brittle star	2.00	10.90	0.2	0.67	0.3	0.01
35	Polychaeta	2.00	8.74	1	3.33	0.3	0.10
36	Hermit crab	2.60	0.02	3	10	0.3	0.15
37	Mysid	2.10	0.01	45	150	0.3	0.44
38	Zooplankton	2.00	5.82	46.21	154	0.3	0.24
39	Phytoplankton	1.00	28.06	64.21			0.50
40	Detritus	1.00	43.00				0.03

Input values are shown in bold and estimated values by Ecopath in normal texts

TL trophic level, *B* biomass (t/km²), *P/B* production/biomass (/year), *Q/B* consumption/biomass (/year), *EE* ecotrophic efficiency.

Fishery Cooperative Association 1987; 2015). The data used in this study encompassed six fishing gear types or fishing methods (i.e., set net, purse seine, bottom trawl, gill net, dredge, and angling including longline fishing) commonly used in the CAK up to a depth of 240 m.

Calculations of *Q/B* for fish functional groups are presented in Online Resource 5. The biomass of each fishery functional group in CAK was estimated by dividing the catch in CAK by the estimated harvest rate from stock assessment reports from the Japan Fisheries Agency and literature (Online Resource 6, Table S6-1) with the assumption that harvest rate is invariable among subareas. Recreational fishing catch statistics could not be included in this study because of limited data availability. The

annual biomass of detritus, mysids, and zooplankton was obtained from literature (Yang 2001; Akiyama et al. 2015; Watari 2015) (Online Resource 6, Table S6-2). The biomass of phytoplankton, the food source for both models, was estimated by the model because of insufficient survey data to estimate it. The precision of *DC* was low among the parameters; therefore, *DC* was adjusted to achieve mass balance in this study after referring to the literature (Table 2 and Online Resource 7, Table S7). In the diet composition table (Table 2), the sum of prey and imports that cover each functional group should be 1 (100%) at the time of model creation. However, because many large fish species temporarily migrate into the model area from outside the model system of CAK, they consume a large

Table 2 Diet composition (DC) and import ratio of functional groups in 1985 and 2013

Prey\predator	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	
1 Sardine				0.01		0.01	0.01	0.01	0.01	0.01		0.01				0.01					0.0001																		
2 Round herring						5E-04				0.0005																													
3 Anchovy				0.1	0.10	0.102	0.11	0.1	0.1	0.1	0.01	0.25	0.01	0.1		0.16	0.1	0.1	0.10	0.05	0.001	0.001	0.001	0.001															
4 Mackerel								0	0	0.001		0.01				0.001					0.001																		
5 Jack mackerel				0		0.003		0.01	0.01	0.05		0.05				0.01	0.02				0.01																		
6 Yellowtail																																							
7 Frigate tuna								0.01	0.01																														
8 Tuna																																							
9 Sailfish																																							
10 Spanish mackerel							0.1		0.1	0.1																													
11 Stingray																																							
12 Flounder																																							
13 Black porgy																																							
14 Red seabream																																							
15 Tilefish																																							
16 Seabass																																							
17 Flying fish										0.001																													
18 Rockfish																					0.0001	0.0001																	
19 Barracuda																					0.001																		
20 Flying squid						0.05	0.05	0.05	0.05			0.01	0.01			0.001	0.05	0.1			0.01																		
21 Other squids					0.05	0.15	0.15	0.05	0.05	0.1		0.1	0.1	0.1	0.05	0.05	0.1	0.10																					
22 Octopus												0.01	0.01																										
23 Ivory shell																																							
24 Sea cucumber																																							

Table 2 (continued)

Prey\predator	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38		
25 Other conch												0.05	0									0.011	0.1															0.02		
26 Crab														0.03	0.05																									
27 Prawn														0	0							0.0005	0.0005	0.0005																
28 Bivalve											0.1	0.2	0.01	0.01								0.1	0.1									0.1	0.1							
29 Dragonet											0.01					0.005																								
30 Tongue sole											0					0.0001																								
31 Goby											0.05					0.01																								
32 Shrimp											0.1	0.01	0.1	0.14	0.002							0.005	0.005																	
33 Starfish													0.01																											
34 Brittle star															0.1																						0.1			
35 Poly-chaeta											0.41	0.35	0.3	0.3	0.3	0.3		0.19																					0.4	
36 Hermit crab											0.01		0.01		0.01	0.01																								
37 Mysid										0.01		0.13	0.1	0.1		0.1																								
38 Zoo-plank-ton	0.6	0.6	0.6	0.48	0.5	0.176	0.12	0.03	0.04	0.233	0.02	0.2	0.1	0.2	0.33	0.331	1	0.35	0.5	0.54		0.669	0.287					0.2	0.2	0.5	0.2	0.6	0.1						0.2	0.1
39 Phyto-plank-ton																																								
40 Detritus											0.2																													
Import	0.4	0.4	0.4	0.41	0.4	0.509	0.56	0.55	0.63	0.545																														
Sum	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	

amount of prey outside the system. The "import" function of Ecopath compensates for such prey consumption outside the system. In this study, the "import" of the migrated large predatory fish was determined by adjusting the EE of the prey to fall within the appropriate range for each functional group, without changing the P/B and Q/B for each functional group.

Ecopath quality and pre-balance diagnostics

The input data quality ratings ("pedigrees") for each model parameter are shown in Online Resource 8. The uncertainty in each of the input parameters (B , P/B , Q/B , and DC) for each functional group in the CAK was assessed by a data grading procedure (Online Resource 8), which provided an error range from $\pm 10\%$ (low uncertainty) to $\pm 90\%$ (high uncertainty) to each parameter, classified according to Aydin et al. (2007). To balance the model, the EE of the functional groups should be lower than one (Christensen et al. 2008). The EE values of the 1985 and 2013 models were compared for each functional group. In addition, pre-balance (PREBAL) diagnostics were run in Ecopath, showing how biomass changes with trophic level. The slope of biomass (log-scaled) across all functional groups arrayed by trophic level should decline by 5–10% (Link 2010; Heymans et al. 2016). The ranges of P/Q values for all functional groups in both models were compared with common values from previous Ecopath literature (Christensen et al. 2005; Watari 2015; Watari et al. 2019). In addition, the TL values of the squid in these models were compared with standard values. The CAK models were checked based on PREBAL comprising 17 criteria, and the results were ranked into three categories: "good," "acceptable," and "caution" (Link 2010).

Ecosystem network analysis for quantifying the effects of the regime shift on the ecosystem

Ecological network analysis outputs several useful indices for comparing ecosystem models. Trophic levels, omnivory indices, mixed trophic impact (MTI), and classification tree were compared between different regimes. The outputs of these models could quantify differences at several levels (between functional groups and between fisheries) between the 1985 and 2013 models to evaluate the effects of the regime shift.

Trophic level (TL)

The trophic position of a functional group in a given ecosystem can be calculated using the following formula (Odum and Heald 1975; Adams et al. 1983):

$$TL_i = 1 + \sum_{j=1}^n (DC_{ji} \times TL_j) \quad (2)$$

where TL is the trophic level of a functional group, with 1 representing the lowest trophic level that includes detritus and the highest trophic level representing the top predators; DC_{ji} is the diet composition representing how much the prey functional group i contributes to the diet of a predator functional group j .

Omnivory index (OI)

The omnivory index (OI) represents the variety in trophic levels for a consumer and is calculated as follows (Libralato 2008):

$$OI_j = \sum_{i=1}^n (DC_{ji} \times (TL_i - (TL_j - 1))^2) \quad (3)$$

The OI of specialist functional groups is close to zero, while the OI of generalist functional groups is close to one (Christensen and Pauly 1992). The system omnivory index (SOI) is a measure of omnivory in a food web. The SOI is an indicator of how prey–predator interactions are distributed among trophic levels. Because it represents the overall complexity of a given ecological network, the comparison of SOI between ecosystems allows the assessment of ecosystem developmental stage and maturity and their respective capabilities to respond to natural and anthropogenic perturbations (Libralato 2008). Temporal changes in SOI mark temporal changes in the ecosystem trophic structure due to fishery activities and environmental changes (Trites et al. 1999). The SOI of a given trophic network is quantified as the weighted averaged OI of all consumers in the network (Christensen and Walters 2004; Libralato 2008):

$$SOI = \frac{\sum_{i=1}^n (OI_i \times \log(Q_i))}{\sum_{i=1}^n \log(Q_i)} \quad (4)$$

where the weighting factor is the logarithm of each consumer's food intake (Q_i).

Mixed trophic impact analysis

MTI analysis was used to evaluate the impacts (m_{ji}) of the change in biomass of one functional group on the biomass of all other functional groups within the system and was calculated according to a previous study (Ulanowicz and Puccia 1990). The interaction between impacting group j and impacted group i is as follows:

$$m_{ji} = DC_{ji} - FC_{ij} \quad (5)$$

where DC_{ji} is the diet composition that represents the contribution of the prey functional group i to the diet of the predator functional group j . FC_{ij} represents the proportion of predation on functional group i by j . In this analysis, fishing gear was included as a predator (Christensen et al. 2008).

Application of the classification tree

In addition, the biomass (B_j) and overall trophic impact (ε_j) of each functional group j were classified into five ecological categories, following the methods of Valls et al. (2015): keystone (category A), high-impact/high-biomass (category B), low-impact/low-biomass (category C), low-impact/high-biomass (category D), and intermediate (category E). The overall trophic impact is calculated as follows:

$$\varepsilon_j = \sqrt{\sum_{i \neq j} m_{ji}^2} \quad (6)$$

where ε_i excludes the MTI of functional groups on themselves and dead groups (i.e., detritus).

Ecosystem network analysis for quantifying the effects of fishing on the ecosystem

The following indicators quantitatively examine the impact of a particular fishery type on a functional group, thereby supporting the identification of fishery types that are suitable for sustainable fishery resource management.

Mean trophic level of catch

The mean trophic level of catch (TL_c) indicates the trophic position of the overall annual catch in a model area (Table 3) and is defined as follows (Lindeman 1942; Odum and Heald 1975):

$$TL_c = \frac{\sum_i (TL_i \times Y_i)}{\sum_i Y_i} \quad (7)$$

where TL_i is the trophic level value of functional group i and Y_i is the annual catch rate of functional group i .

The Ecopath model of the CAK was built using indicators related to Odum's theory of ecosystem development (Odum 1969). Several ecosystem models worldwide, including those of the CAK, characterized by model year, model size (km²), number of functional groups, model depth range, Köppen climate classification, indicators, TL_c , total catch, and SOI were compared (Table 3, Table 4).

Primary production required

Two indices were considered to assess the impact of commercial fisheries on ecosystems. The primary production required (PPR) was estimated to evaluate the amount of primary production necessary to support the production of fishery catch (Pauly and Christensen 1995). The PPR of a functional group i (PPR_i) is calculated as follows:

$$PPR_i = \sum_i \sum_{paths} \left(Y_i \times \prod_{predator, prey} \frac{Q_{predator}}{P_{predator}} \times DC'_{predator, prey} \right) \quad (8)$$

where Y_i is the annual catch rate of the functional group i , $P_{predator}$ is the production of the predator, $Q_{predator}$ is the consumption of the predator, and $DC'_{predator, prey}$ is the percentage of the prey in the diet composition of the predator. The PPR in the system was calculated as

$$PPR = \sum PPR_i \quad (9)$$

In addition, the PPR per total primary production (% PPR) was calculated for each model as follows:

$$\%PPR = PPR / \text{Total Primary production} \quad (10)$$

Index of the loss in secondary production due to fisheries catch

The index of the loss in secondary production due to fisheries catch ($Lindex$) was calculated using PPR (Libralato et al. 2008). A larger value of $Lindex$ indicates a tendency toward overfishing (Coll et al. 2008). The value of $Lindex$ ranges from 0 to 0.25, and is calculated as:

$$Lindex = -\frac{1}{P_1 \cdot \ln TE} \times \sum_i^m (PPR_i \times TE^{TL_i-1}) \quad (11)$$

where P_1 is the total primary production, TE is the transfer efficiency, PPR_i is the primary production required for the functional group i ($i = 1, \dots, m$), and TL_i is the trophic level of the functional group i .

Classification as sustainably fished

The probability for the ecosystem to be classified as sustainably fished (P_{sust}) was estimated to evaluate the relationship between the ecosystem and fisheries using the $Lindex$ (Coll et al. 2008). P_{sust} values were used to assess the sustainability of fishing using the 1985 and 2013 models in CAK. P_{sust} is based on the definition of overfishing as stated by

Table 3 Ecosystem details of models in the world

Country	Model systems	Model year	Size (km ²)	Number of functional groups	Depth (m)	Köppen climate classification	Mean trophic level of the catch	Total catch (t/ km ² /year)	SOI	Reference
Japan	Tango Bay	1985	2230	40	0–240	Cfa	3.06	21.5	0.29	This study
		2013	2230	40	0–240	Cfa	3.20	4.0	0.36	
	Seto Inland Sea	2001–2005	3100	19	0–60	Cfa	3.22	7.5	0.12	Watari et al. (2015)
	Northeastern Japan	1997–2009	14,344	46	100–1000	Cfa	3.25	4.1	–	Yonezaki et al. (2016)
China	North Pacific	2013	913,102	41	0–1000	Cfa	3.20	1.7	0.10	Watari et al. (2019)
	East China Sea	2000	570,000	21	0–200	Cfa	3.32	8.0	–	Cheng et al. (2009)
	Ionian Sea	1964	1021	19	0–200	Cs	3.10	1.1	0.10	Piroddi et al. (2010)
Tunisia	Gulf of Gabes	2000's	35,900	41	20–200	Cfb	3.44	1.7	–	Hattab et al. (2013)
Spain	Catalan deep open slope	2009	850	20	1000–1400	Cs	–	no catch	0.29	Tecchio et al. (2013)
France	Cantabrian Sea	1994	16,000	28	0–100	Cfb	3.66	9.4	0.27	Sánchez and Olaso (2004)
	French Continental Shelf	1994	102,585	32	0–200	Cfb	–	1.3	0.21	Lassalle et al. (2012)
	North Sea	1991	570,000	68	0–400	Cfb	3.60	4.9	0.23	Mackinson and Daskalov (2007)
Canada	Central Baltic Sea	1974	240,669	22	0–401	Df	–	3.0	–	Tomczak et al. (2012)
	Straits of Georgia, B.C	1960	6800	38	0–447	Cfb	3.51	7.1	–	Preikshot et al. (2013)
USA	Gulf of Mexico	2009	145,000	47	0–200	Cfa	2.60	4.0	0.19	Geers et al. (2016)
China	The Pearl River (mangrove)	2007	17,200	14	0–2.5	Am	–	1.8	0.19	Xu et al. (2011)
Australia	Tasmania Island	2000's	137,000	47	0–40	Cfb	3.09	0.1	0.46	Watson et al. (2013)
Ethiopia	Lake Hayq (fresh water)	2007	23	15	0–88	Cfa	2.46	–	0.22	Fetahi et al. (2011)

Year, size (km²), number of functional groups, depth (m), Köppen climate classification, mean trophic level of the catch, total catch (t/km² year), system omnivory index (SOI), and references of models are shown. Empty cells represent missing data. All above models are mass-balanced and proofed by checking values of ecotrophic efficiency (EE<1)

*Used PREBAL for checking model consistency

Table 4 Comparison of values of %PPR and catch between 1985 and 2013 model in CAK for each fisheries functional group

	Group name	PPR/TotPP (%)		Di*	Catch(t/km ² /year)		Omnivory Index	
		1985	2013		1985	2013	1985	2013
1	Sardine	5.826	0.631	-5.195	15.76	0.29	0.13	0.38
2	Round herring	0.018	0.002	-0.016	0.11	0.002	0.13	0.38
3	Anchovy	0.332	1.026	0.694	1.96	1.02	0.13	0.38
4	Mackerel	4.003	0.113	-3.890	1.22	0.05	0.58	0.57
5	Jack mackerel	0.080	0.325	0.245	0.44	0.32	0.53	0.53
6	Yellowtail	1.025	4.124	3.099	0.31	0.55	0.16	1.14
7	Frigate tuna	0.603	0.369	-0.234	0.17	0.04	0.54	1.14
8	Tuna	0.495	0.435	-0.060	0.06	0.02	0.10	1.34
9	Sailfish	0.111	0.108	-0.003	0.01	0.01	0.41	1.46
10	Spanish mackerel	0.477	3.835	3.358	0.04	0.77	0.37	0.96
11	Stingray	0.081	0.000	-0.081	0.004	0.000005	0.53	0.38
12	Flounder	0.058	0.059	0.001	0.03	0.02	0.59	0.22
13	Black porgy	0.168	0.134	-0.034	0.004	0.011	0.33	0.10
14	Red seabream	0.210	0.360	0.150	0.02	0.03	0.24	0.16
15	Tilefish	0.032	0.180	0.148	0.002	0.01	0.45	0.22
16	Seabass	0.045	0.755	0.710	0.01	0.05	0.47	0.19
17	Flying fish	0.177	0.279	0.102	0.50	0.10	0	0
18	Rockfish	0.236	0.158	-0.078	0.03	0.02	0.68	0.48
19	Barracuda	3.814	1.949	-1.865	0.36	0.19	0.68	0.52
20	Flying squid	0.090	0.150	0.060	0.01	0.06	0.82	0.48
21	Other squids	0.827	0.485	-0.342	0.23	0.22	1.02	0.29
22	Octopus	0.117	0.044	-0.073	0.04	0.03	0.71	0.27
23	Ivory shell	0	0.000002	0.000002	0	0.00001	0.13	0.20
24	Sea cucumber	0.004	0.005	0.001	0.12	0.05	0	0
25	Other conch	0.001	0.003	0.002	0.01	0.02	0	0
26	Crab	0.021	0.002	-0.019	0.05	0.002	0.21	0.21
27	Prawn	0.005	0.004	-0.001	0.02	0.004	0.16	0.24
28	Bivalve	0.001	0.039	0.039	0.02	0.10	0	0.16
	Total	18.86	15.57		21.47	4.00		

*Di is difference of %PPR between 1985 and 2013

Murawski (2000). Coll et al. (2008) categorized 51 Ecopath models into two groups: overfished (Group 1) and sustainably fished (Group 2). P_{sust} is expressed as:

$$P_{sust}(L^*) = \frac{P(L_2 > L^*)}{P(L_2 > L^*) + P(L_1 < L^*)} \quad (12)$$

where L^* is the value of L_{index} of the 1985 or 2013 models in the CAK and $P(L_1 < L^*)$ is the number of cases within Group 1 that have index values lower than the chosen value L^* . $P(L_2 > L^*)$ is the number of cases within Group 2 that have index values larger than the chosen value of L^* . A high value of P_{sust} ($P_{sust} > 0.75$) represents a sustainably fished ecosystem (Coll et al. 2008).

Environmental variability

The annual average SST in the southwestern Sea of Japan from 1901 to 2019 (SST anomaly: https://www.data.jma.go.jp/kaiyou/data/shindan/a_1/japan_warm/cfig/data/areaE_SST.txt “Accessed 2 Feb 2021”) obtained from the Japan Meteorological Agency and the catch statistics in Kyoto Prefecture (e-Stat: https://www.e-stat.go.jp/stat-search/files?page=1&layout=datalist&toukei=00500216&tstat=000001015174&cycle=0&tclass1=000001034726&stat_infid=000031605852&tclass2val=0 “Accessed 2 Feb 2021”) were used. The annual SST anomaly time-series data were compared with the catch statistics for the six functional groups in the CAK from 1965 to 2015. The cold regime period was

from 1970 to 1986, whereas the period after 1987 was considered the warm regime period in the Sea of Japan (Tian et al. 2013; Yatsu 2019). Thus, the 1985 model belongs to the cold regime and the 2013 model belongs to the warm regime in the CAK.

Results

Mass balancing

The basic input and estimated parameters are presented in Table 1. The diet composition tables were modified to maintain the mass balance of the models (Table 2). The Ecopath models were balanced in both 1985 and 2013. When the CAK Ecopath model could not be balanced using the reference data, the values of diet composition were changed to build a balanced model. Diet compositions of detritivorous functional groups were balanced using detritus dependence as the final adjustment. Owing to drastic changes in the comparative biomass of anchovy and sardine from 1985 to 2013 (Fig. 2b, c, Table 4), the main diet composition for many predators had to be assigned as sardine in 1985 and anchovy in 2013.

Ecopath quality

The details of pedigree are shown in Online Resource 8. The same pedigree values were used for both 1985 and 2013 because the data source was equivalent. The data pedigree for *B* was considered 3 for fishery functional groups and 2 for benthos functional groups. However, that of phytoplankton was considered 8 because it was estimated by Ecopath. The data pedigree for *Y* of fishery functional groups was 2. The data pedigree for *P/B* of all functional groups was considered 7. The data pedigree for *Q/B* was considered 6 for fish functional groups, 7 for cephalopod functional groups, and 8 for benthos functional groups. The data pedigree for *DC* of all functional groups was 6.

The highest *EE* value in the 1985 model was 0.86 (#14 and #38) and in the 2013 model was 0.87 (#5). The ranges of *P/Q* were 0.11–0.30 in 1985 and 0.12–0.30 in 2013. PREBAL diagnostics resulted in either “good” or “acceptable” for all the criteria, indicating that the models were well-balanced. All PREBAL results are presented in Online Resource 9.

Basic outputs and trophic level

Twenty-eight of the evaluated functional groups were targeted by the CAK fisheries (Table 4 and Online Resource 4, Table S4). The total fishery catches of 28 functional groups in the CAK decreased from 21.5 t/km²/year in 1985

to 4.0 t/km²/year in 2013. Particularly, the catch of sardine declined from 15.8 to 0.3 t/km²/year during this period. The prey–predator relationship between all functional groups in CAK is presented in a flow diagram (Fig. 3). The mean trophic level of the CAK catches increased from 3.06 to 3.20 between the two years (Table 3). In 1985 and 2013, benthic invertebrates presented lower trophic levels, ranging from 2.00 to 3.47, mainly feeding on detritus and other benthic organisms (Table 1). The trophic level of demersal fish ranged from 2.55 to 3.49 and pelagic functional groups presented higher trophic levels ranging from 3.00 to 4.08. In the CAK, the tuna (#8) functional group represented by *Thunnus orientalis* and the sailfish (#9) functional group, including *Istiophorus platypterus* and *Makaira mazara*, presented the highest trophic level (4.07 and 4.08 in 1985 and 4.08 and 4.07 in 2013, respectively) (Table 1). The *SOI* value in 2013 (0.36) was higher than that in 1985 (0.29) (Table 3). The values of imports for functional groups outside the model area were set between 0.1 and 0.56 for pelagic functional groups (#1, #2, #3, #4, #5, #6, #7, #8, #9, #10, #20, and #21) because of wide range migration habitats. In 2013, the model could not be balanced unless anchovy, which had the largest biomass among the small pelagic fishes, was set as the main prey for piscivorous animals (#4, #5, #6, #7, #8, #9, #10, #11, #12, #13, #14, #16, #18, #19, #20, and #21).

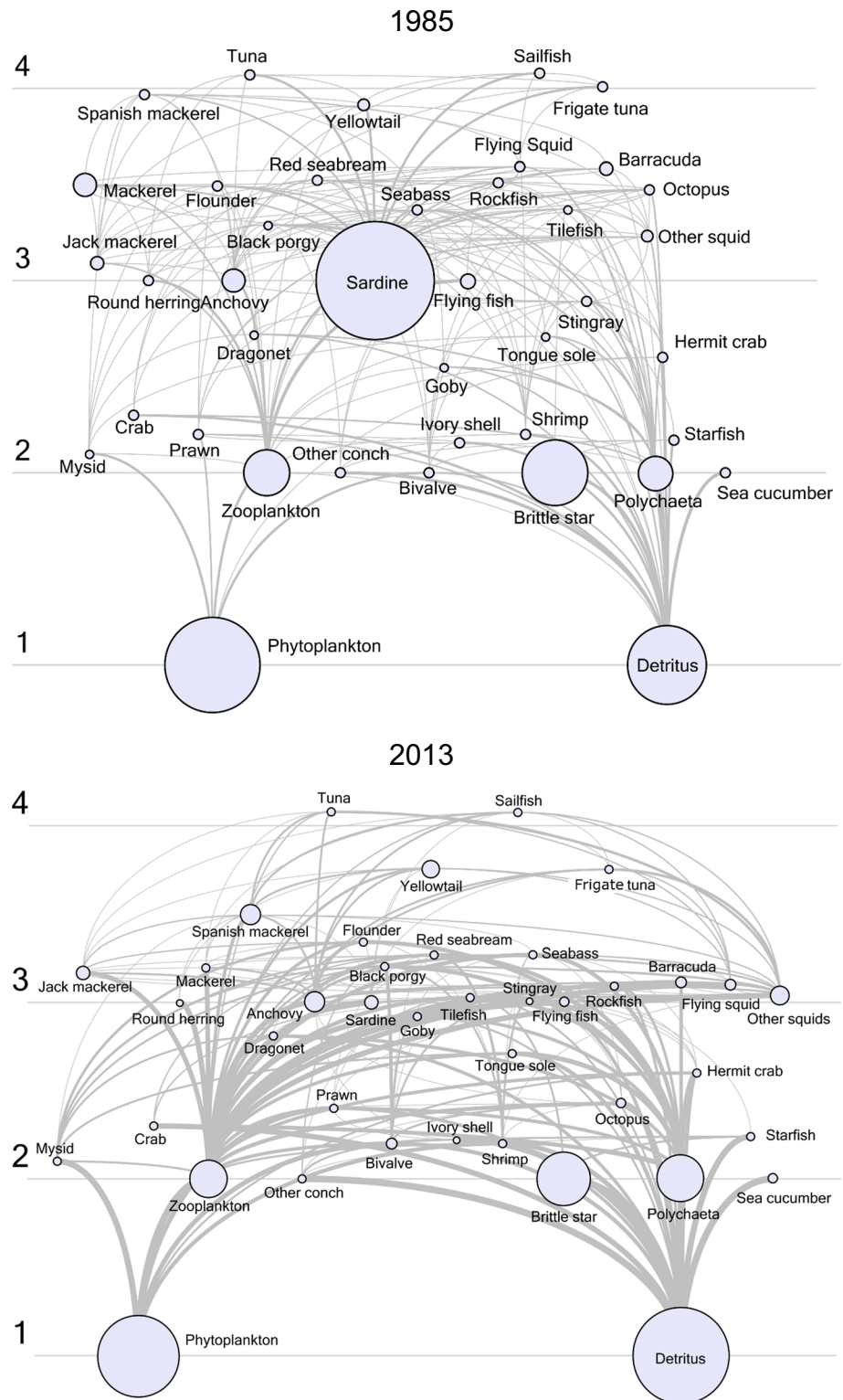
Mixed trophic impact analysis

The MTI results showed that the sardine (#1) had a high impact on many functional groups in 1985, but the impact was low in 2013 (Fig. 4). In contrast, the impacts of yellowtail (#6), Spanish mackerel (#10), and seabass (#16) increased from 1985 to 2013. Set net fisheries had negative impacts on pelagic functional groups but positive impacts on benthic functional groups in both 1985 and 2013. There were many negative impacts of predator functional groups (impacting groups) on prey functional groups (impacted groups) as a direct prey–predator relationship. The purse seine fisheries caught 25% of the mackerel annual biomass and had the most significant negative impact on mackerel in 1985. However, it had minimal impact on other species, including sardine, affecting less than 7% of the sardine annual biomass (Table 1 and Online Resource 4, Table S4).

Application of the classification tree

In the classification tree of functional groups, data from the two years showed a similar tendency: distributed from the lower left to the top right in Fig. 5. In the 1985 model, the stingray (#11), black porgy (#13), and seabass (#16) functional groups were classified as keystone species in category (A), but there were no keystone species in 2013. Keystone species have a low biomass and high trophic

Fig. 3 Flow diagram of the CAK in 1985 and 2013. Circle sizes indicate biomass (on a log scale) of functional groups. Widths of the lines show magnitudes of flow between prey and predator functional groups. Numbers on the left side indicate the trophic levels. Due to the limitations of the Ecopath software function, the relationship between biomass and circle size differs between the two years



impact. Significant differences between the two years were observed: decrease in the biomass and trophic impact of the sardine (#1), stingray (#11), mackerel (#4), and ivory shell (#23) functional groups and increase in the biomass and trophic impact of the yellowtail (#6), tilefish (#15),

bivalve (#28), dragonet (#29), tongue sole (#30), and goby (#31) functional groups from 1985 to 2013. There were only the flying fish (#17) functional group in category (D) in 1985 but no functional groups in this category in 2013.

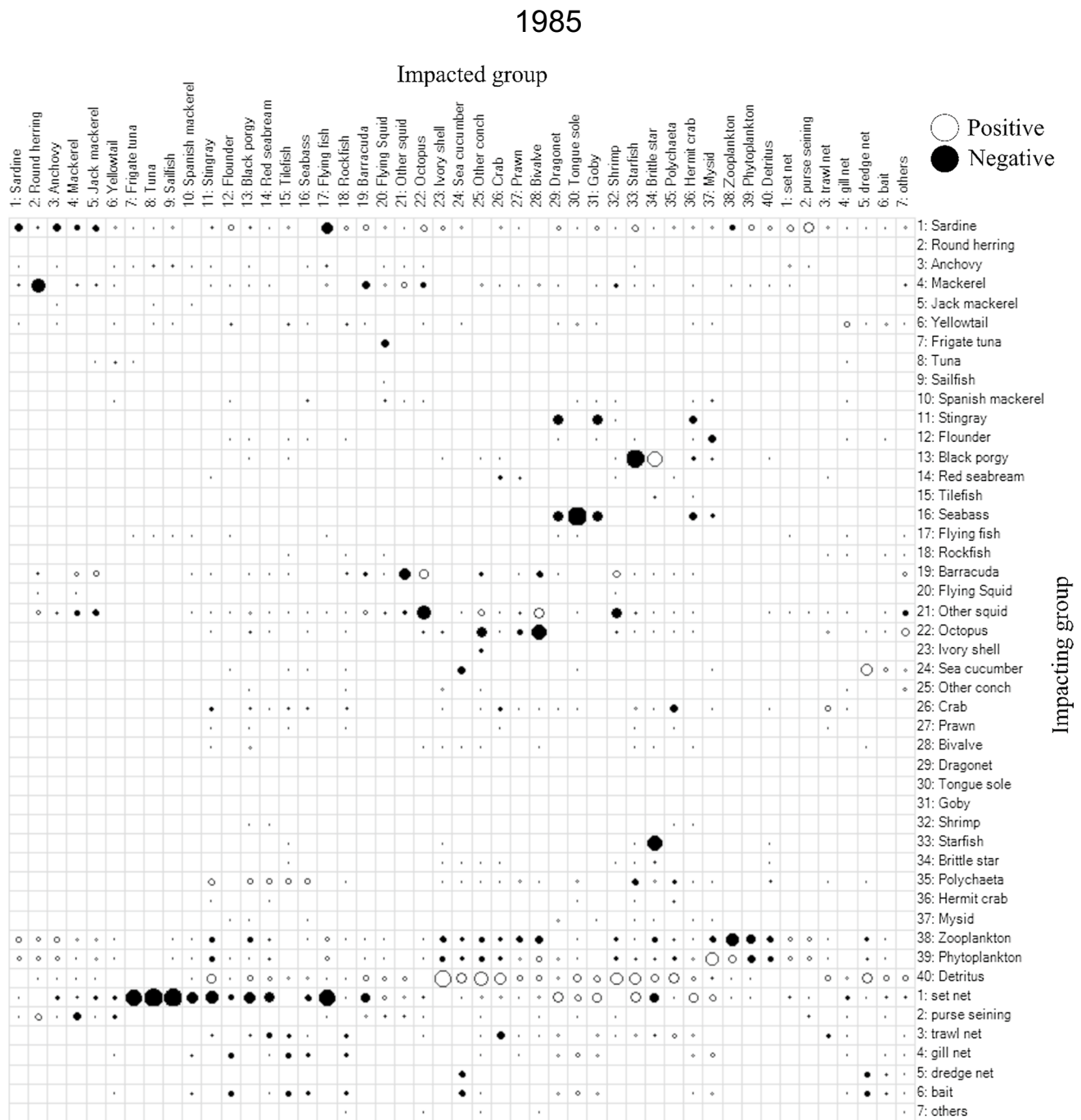


Fig. 4 Matrix of total mixed trophic impact (MTI) in 1985 and 2013. Black filled circles show negative impacts and unfilled circles show positive impacts

Most of the CAK functional groups fell into the intermediate category (E) in both years.

Index for quantifying the effects of fishing on the ecosystem

The value of %PPR in the CAK was higher in 1985 (18.86%) than in 2013 (15.57%), indicating that the fishery

catch in 1985 demanded more primary production than that in 2013 (Table 4). In 1985, sardines had the highest functional group-specific %PPR (5.83%), followed by mackerel (4.00%). However, the %PPR of the two groups declined substantially in 2013. By contrast, the %PPR of yellowtail and Spanish mackerel increased significantly in 2013 (4.12% and 3.84%, respectively).

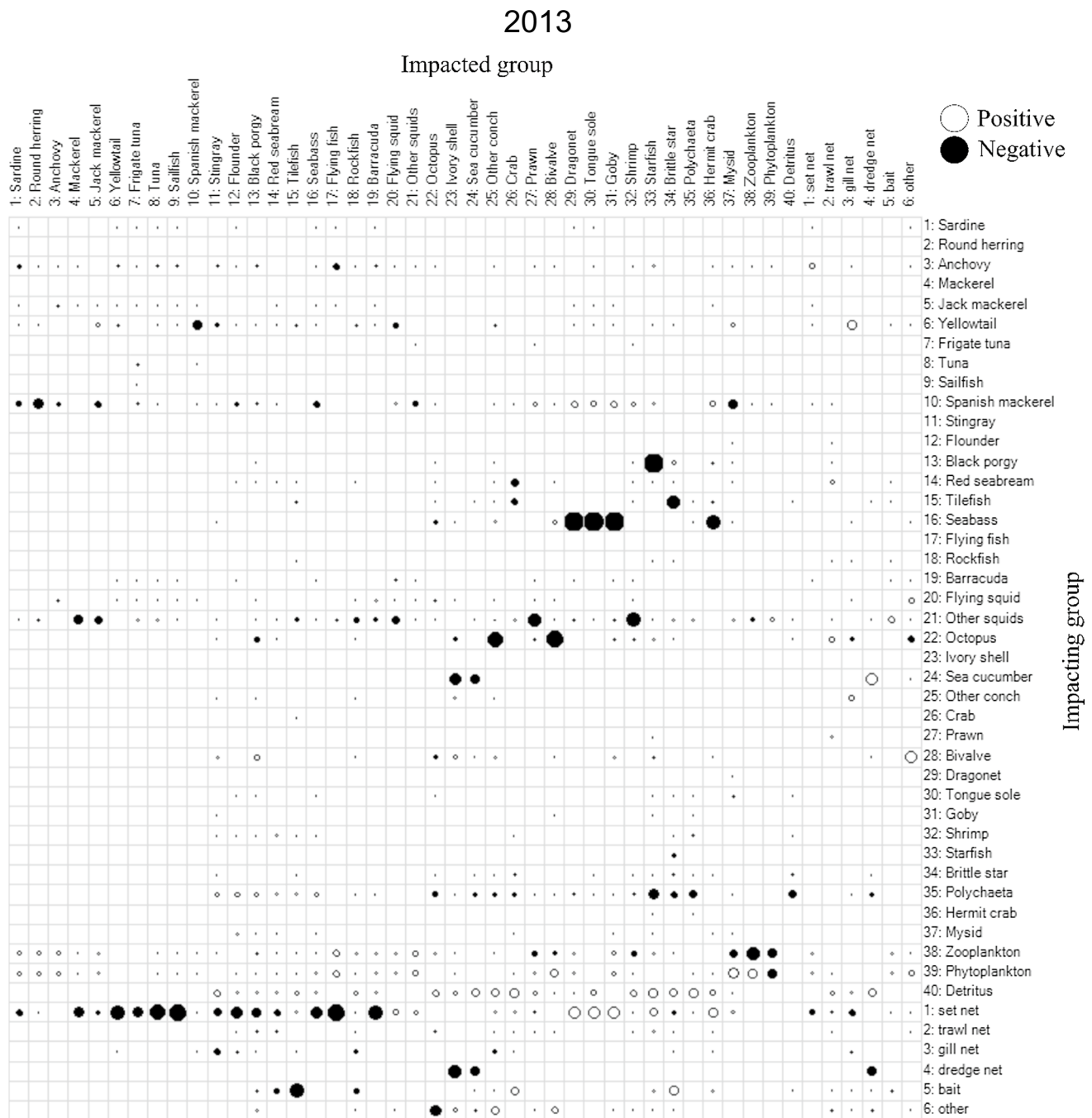


Fig. 4 (continued)

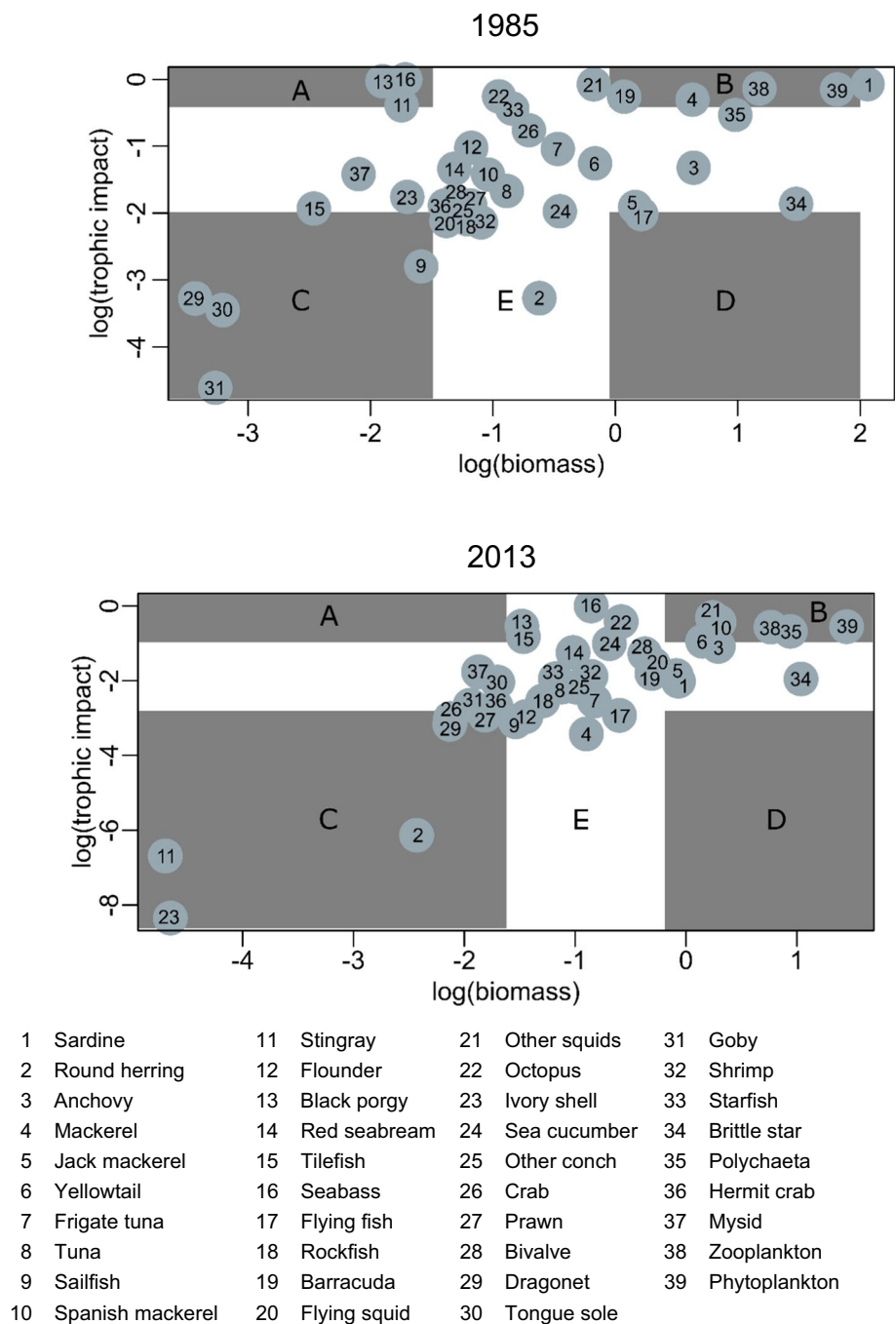
The value of *Lindex*, which is an index for overfishing, was greater in 1985 (0.596) than in 2013 (0.028). P_{sust} in 1985 (37.1%) was lower than in 2013 (72.1%).

Environmental variability

The average annual SST anomaly in the cold regime period (from 1970 to 1986) was -0.29°C and that in the

warm regime period (from 1988 to 2015) was 0.16°C . The annual SST anomaly trend increased from 1964 to 2015 (Fig. 2a).

Fig. 5 Application of the classification tree according to Valls et al. (2015) to the CAK food web in 1985 and 2013. The scatterplot of functional groups shows the log-transformed biomass on the x-axis, and the log-transformed trophic impact (in squared values; ϵ_i^2) on the y-axis. Each point is a functional group in the model, identified with a group number and a group name (indicated in the legend). The five group categories are keystone (category A), high-impact and high-biomass (category B), low-impact and low-biomass (category C), low-impact and high-biomass (category D), and intermediate (category E). The five categories were defined by the first quartile (Q1) and third quartile (Q3) for both the log-transformed biomass and the log-transformed trophic impact



Discussion

Model construction

An Ecopath model must be balanced to analyze the ecosystem network properly. The two CAK Ecopath models of 1985 and 2013 were successfully built based on fisheries and survey data of 40 functional groups. To tune the model, the relationships between prey and predator functional groups, the catch and biomass of functional groups,

the inflow and outflow of detritus, and the import and export of functional groups must be balanced. The range of P/Q was 0.11–0.30 in 1985 and 0.12–0.30 in 2013, which are reliable P/Q values according to Christensen et al. (2005). The indices used to evaluate the quality of the model, such as EE values for all functional groups and the values of P/Q and TL for squids in both the 1985 and 2013 models, were within acceptable ranges (Christensen et al. 2008). Therefore, the outputs of these models were considered reliable.

The biomass of many functional groups was estimated from the stock assessment results or benthos survey. P/B and Q/B are similar to the values used for closely related species in the CAK. Although these values need to be improved in the future, the values of closely related species can be used as the P/B and Q/B of the Ecopath input values for practical purposes. The precision of biomass and EE estimates from Ecopath is equivalent to the precision of the input data (Essington 2007). In this study, the accuracy of biomass for benthos functional groups was high because the data were collected through sampling in the field. Important error sources in commercial fish statistics are thought to originate from discarded fish from fishing boats (Akiyama 2007) and catches by recreational fishing (Akiyama 2007; Terashima et al. 2020). We considered the present statistical data to be reliable because EE values of most of the functional groups in the present study were similar to those in the Pacific area of Japan (Watari et al. 2019). The two main problems with Ecopath as an ecosystem model are errors due to input values and the treatment of migratory fish from outside the system, both of which are discussed below.

According to the literature, 14 (#1, #2, #3, #4, #5, #6, #7, #8, #9, #10, #11, #17, #19, and #20) out of 28 commercially important functional groups in CAK are considered migrants, which utilize this area during their seasonal migration. Because CAK is only a small part of the migratory range of these migratory functional groups, a part of their catch was set as imports in Ecopath. The migration routes of these fishes are strongly affected by changes in ocean circulation and currents, resulting in high annual variability in the influx of migratory functional groups into the CAK. Thus, the pedigree of many migratory fishes is considered low-ranked. Tian et al. (2012) showed that migratory fishes determine their migration routes in response to seawater temperature; Ecospace can consider spatial and temporal change of mass balance and could thus be used to incorporate environmental factors and the migration routes of migratory functional groups into the simulations, resulting in a more accurate prediction (Pauly et al. 2000).

Fisheries ecosystem analysis

The major change in catches in the CAK between 1985 and 2013 was a drastic decline in sardine and an increase in Spanish mackerel in 2013. The total fishery catches of 28 functional groups in the CAK decreased from 21.5 t/km²/year in 1985 to 4.0 t/km²/year in 2013 mainly because of the decline in sardine (Table 4). In addition, other small and medium-sized pelagic fishes, such as anchovy, round herring, and mackerel, tended to decline in 2013, while large predatory fishes such as yellowtail and sea bass increased. The Ecopath outputs reflected these changes in fish species composition.

The TL_c and SOI increased from 3.06 to 3.20 and 0.29 to 0.36 between 1985 and 2013 (Table 3), respectively. These changes were mainly due to the sharp decline in sardine biomass ($TL=3.00$ in both 1985 and 2013) and the increase in high trophic-level functional groups, such as Spanish mackerel ($TL=3.97$ in 1985 and 3.49 in 2013) and yellowtail ($TL=3.91$ in 1985 and 3.76 in 2013) in 2013. These changes in SOI before and after the regime shift are consistent with the results of Tian et al. (2014) for the Tsushima Warm Current. The 2013 total catch per unit area of CAK was similar to that of other areas (Table 3), but less than one-fifth of that in 1985, indicating that sardine had a high impact on the ecosystem and food web. Large-sized piscivorous fishes, including Spanish mackerel and yellowtail, have a wide distribution from southwest to northeast, and the increase in water temperature has positive effects on their biomass in the Sea of Japan (Tameishi et al. 2005; Tian et al. 2012; Tojima et al. 2013). In CAK too, their catch increased gradually from the 1990s with increasing SST (Fig. 2a, e, f, g). The results of the MTI showed that the impact of yellowtail, Spanish mackerel, and seabass on other small functional groups increased from 1985 to 2013 (Fig. 4).

In the 1985 classification tree analysis, three functional groups, namely stingray (#11), black porgy (#13), and seabass (#16), were classified as keystone species (category A, Fig. 5). A keystone species is defined as one that has a high trophic impact even at a low total biomass (Valls et al. 2015). However, in 2013, the biomass of black porgy and seabass increased and they shifted to the intermediate group (category E) and stingray shifted to a low-impact and low-biomass group, leading to an absence of keystone species. The loss of keystone species in an ecosystem changes community architecture (Paine 1995) and indicates ecosystem shifting (Whitrod and Zukowski 2019). However, it is difficult to understand the relationship between the presence or absence of keystone species and ecosystem stability at the present stage, because various definitions and methods to identify keystone species have been used (Valles et al. 2015) and the roles and mechanisms of keystone species vary depending on the type of ecosystem (Bond 2001).

Comparing the impacts of fisheries on ecosystems between 1985 and 2013, the %PPR decreased from 18.86 to 15.57%, the $Lindex$ decreased from 0.596 to 0.028, and P_{sust} , the criterion for sustainable fishery (Coll et al. 2008), increased from 37.1 to 72.1%. Higher P_{sust} in 2013 is thought to be caused by the lower $Lindex$ being related to the increased TL_c in 2013 fisheries, which caught less sardine and more Spanish mackerel and yellowtail. These changes in fishery indices show that the relationship between the coastal ecosystem and fisheries in the CAK shifted to a more sustainable state from 1985 to 2013. The increase in SOI in 2013 also supported the idea that the change from

an ecosystem dominated by mid-trophic sardine to one comprising a diverse food web led to increased ecosystem stability.

The set net is a balanced harvest fishery method (Nédélec and Prado 1990; Kolding et al. 2016; Kiyota et al. 2020) that has been suggested to be effective for the sustainable use of fishery resources (Garcia et al. 2016). It enables multi-species management by changing the mesh size, season, and location. Although there has been some technological innovation in set net fisheries, P_{sust} values indicate that the fishery structure, mainly based on set net fisheries, remains sustainable in terms of the ecosystem effects of fishing, even in recent years. This result can be further evaluated by applying an entire ecosystem model. Although there are limitations in the used models, Ecopath, which assess the effects of complex factors on fishery resources in a complete ecosystem, is helpful for achieving sustainable fisheries. Monitoring the environment, fishery resource status, and ecosystems is vital for achieving a high modeling precision.

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Declarations

Conflict of interest The authors declare that they have no conflicts of interest.

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